

Increasing the genome-targeting scope and precision of base editing with engineered Cas9-cytidine deaminase fusions

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Base editing induces single-nucleotide changes in the DNA of living cells using a fusion protein containing a catalytically defective *Streptococcus pyogenes* Cas9, a cytidine deaminase, and an inhibitor of base excision repair¹. This genome editing approach has the advantage that it does not require formation of double-stranded DNA breaks or provision of a donor DNA template. Here we report the development of five C to T (or G to A) base editors that use natural and engineered Cas9 variants with different protospacer-adjacent motif (PAM) specificities to expand the number of sites that can be targeted by base editing 2.5-fold. Additionally, we engineered base editors containing mutated cytidine deaminase domains that narrow the width of the editing window from ~5 nucleotides to as little as 1–2 nucleotides. We thereby enabled discrimination of neighboring C nucleotides, which would otherwise be edited with similar efficiency, and doubled the number of disease-associated target Cs able to be corrected preferentially over nearby non-target Cs.

In most genome editing applications using CRISPR–Cas9 (ref. 2), the Cas9 nuclease, complexed with a single guide RNA (sgRNA), induces a double-stranded DNA break (DSB) at a target site specified by the sgRNA sequence. In the presence of a donor DNA template homologous to the sequences flanking the DSB, gene correction can be achieved through homology-directed repair^{3,4}. Under most non-perturbative conditions, however, this type of correction is inefficient and dependent on cell state and cell type. Instead, DSBs are preferentially repaired by the alternative pathway of non-homologous end-joining, which results in stochastic insertions and deletions (indels)^{3,4}. As most of the known genetic variations associated with human disease are point mutations⁵, methods that can more efficiently make precise point mutations are needed.

We recently described base editing, which enables replacement of a target base pair with a different base pair in a programmable manner without inducing DSBs¹. Our first examples of base editors are fusion proteins comprising a catalytically inactivated (dCas9)

or nickase form of *Streptococcus pyogenes* Cas9 (SpCas9), a cytidine deaminase such as APOBEC1, and an inhibitor of base excision repair such as uracil glycosylase inhibitor (UGI) to convert cytidines into uridines within a five-nucleotide window specified by the sgRNA¹. Our third-generation base editor, BE3, converts C:G base pairs to T:A base pairs, including disease-relevant point mutations, in multiple cell lines with higher efficiency and lower indel frequency than what can be achieved using other genome editing methods¹. Subsequent studies have validated the base-editing approach in a variety of settings^{6–10}.

Efficient editing by BE3 requires the presence of an NGG PAM that places the target C within a 5-nucleotide window near the PAM-distal end of the protospacer (positions 4–8, counting the PAM as positions 21–23)¹. This PAM requirement limits the number of sites in the human genome that can be efficiently targeted by BE3, as most sites of interest lack an NGG 13–17 nucleotides downstream of the target C. Moreover, BE3 efficiently converts all Cs within the ~5-bp editing window to Ts, which can potentially introduce undesired changes to the target locus¹. Here we report new C:G to T:A base editors that address both of these limitations and thereby substantially expand the targets suitable for base editing.

We hypothesized that any Cas9 homolog that binds DNA and forms an ‘R-loop’ complex¹¹ containing a single-stranded DNA bubble could in principle be converted into a base editor. The Cas9 homolog from *Staphylococcus aureus* (SaCas9) can mediate efficient genome editing in mammalian cells and requires an NNGRRT PAM¹². We replaced the nickase form of SpCas9 with that of SaCas9 in BE3 to generate APOBEC1–SaCas9n–UGI (SaBE3), and transfected HEK293T cells with plasmids encoding SaBE3 and sgRNAs targeting six endogenous human genomic loci (Fig. 1a). After 3 d, we used high-throughput DNA sequencing (HTS) to quantify base-editing efficiency. SaBE3 very efficiently edited target Cs at human genomic loci, with conversion efficiencies of ~50–75% of total DNA sequences converted from C to T, without enrichment for transfected cells (Fig. 1b). The efficiency of SaBE3 on NNGRRT-containing target sites in general exceeded that of BE3 on NGG-containing target sites¹. Perhaps due to greater solvent exposure of the strand not paired with the guide RNA¹³, editing with

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SaBE3 can also result in detectable base editing at target Cs at positions outside of the canonical BE3 activity window (**Fig. 1b**). By comparison, BE3 showed markedly reduced editing for the same non-NGG target sites (0–11% editing), consistent with the PAM requirement of SpCas9 (**Supplementary Fig. 1a**)¹⁴. These data show that SaBE3 can mediate base editing at sites not accessible to BE3.

To further broaden the targeting range of base editors, we used recently engineered Cas9 variants with expanded or altered PAM specificities. Joung and co-workers recently reported SpCas9 mutants that accept NGA (VQR-Cas9), NGAG (EQR-Cas9), or NGCG (VRER-Cas9) PAM sequences¹⁵, as well as an engineered SaCas9 variant containing three mutations (SaKKH-Cas9) that relax the variant's PAM requirement to NNNRRT¹⁶. We replaced the SpCas9 portion of BE3 with these four Cas9 variants to produce VQR-BE3, EQR-BE3, VRER-BE3, and SaKKH-BE3, which should target NGAN, NGAG, NGCG, and NNNRRT PAMs, respectively. We transfected HEK293T cells with plasmids encoding these constructs and sgRNAs targeting several genomic loci for each new base editor, and measured C-to-T base conversions using HTS.

SaKKH-BE3 edited sites with NNNRRT PAMs with efficiencies up to 62% of treated, non-enriched cells (**Fig. 1b,c**). As expected, SaBE3 was unable to efficiently edit targets containing PAMs that were NNHRRT (where H = A, C, or T) (**Fig. 1c**). VQR-BE3, EQR-BE3, and VRER-BE3 exhibited more modest, but still substantial base-editing efficiencies of up to 50% of treated, non-enriched cells at genomic loci with the expected PAM requirements and an editing window similar to that of BE3 (**Fig. 1d–f**). Base-editing efficiencies of VQR-BE3, EQR-BE3, and VRER-BE3 closely paralleled the reported PAM requirements of the corresponding Cas9 nucleases; for example, EQR-BE3 was unable to efficiently edit targets containing NGAH PAM sequences (**Fig. 1e**). As expected, BE3 was unable to efficiently edit sites with NGA or NGCG PAMs (0–3% efficiency) (**Supplementary Fig. 1b**). To confirm that the five new base editors functioned in multiple mammalian cell types, we assessed their performance in U2OS cells and observed robust editing, albeit with slightly lower editing and/or transfection efficiency than in HEK293T cells (**Supplementary Fig. 2a**).

For each of the altered-PAM base editors, we selected two on-target loci that had been previously analyzed for Cas9 off-target cleavage and sequenced seven off-target loci for the SaBE3 constructs and ten off-target loci for the SpBE3 constructs^{15,16} (**Supplementary Fig. 3a**). Consistent with our previous study¹, we detected off-target base editing by SaBE3 and SaKKH-BE3 at a subset of known Cas9 off-target cleavage loci containing an appropriately placed target C (**Supplementary Fig. 3b**). In contrast, we observed substantially less off-target base editing from VQR-BE3 or EQR-BE3 at ten known off-target loci of VQR SpCas9 (ref. 15), suggesting that these base editors may offer enhanced specificity (**Supplementary Fig. 3c**). Collectively, the properties of SaBE3, SaKKH-BE3, VQR-BE3, EQR-BE3, and VRER-BE3 establish that base editors behave in a modular fashion that facilitates our ability to repurpose Cas9 homologs and engineered variants for base editing.

Next we sought to develop base editors with altered activity window widths. All Cs within the 5-nucleotide activity window of BE3 are typically converted to Ts with comparable efficiency¹. The ability to modulate the width of this window is useful when it is important to edit only a subset of Cs present in the activity window, such as cases in which the target C is adjacent to other Cs that, if changed, would cause undesired mutations to the gene of interest.

We previously noticed that the length of the linker between APOBEC1 and dCas9 modulates the number of bases that are acces-

sible by APOBEC1 *in vitro*¹. In HEK293T cells, however, varying the linker length did not significantly modulate the width of the editing window, suggesting that in the cellular milieu, the relative disposition of dCas9 and the cytidine deaminase are not strongly determined by linker length (**Supplementary Fig. 4**). We hypothesized that truncating the 5' end of the sgRNA might narrow the base-editing window by reducing the length of single-stranded DNA accessible to the deaminase. Although for some target loci, truncated guide RNAs with 16- or 17-base spacers showed narrowed editing windows, we observed no consistent changes in the editing window width when using truncated sgRNAs with 15- to 19-base spacers (**Supplementary Fig. 5**).

As an alternative approach, we envisioned that mutations to the deaminase domain might narrow the width of the editing window. Because the high activity of APOBEC1 likely contributes to the deamination of multiple Cs per DNA binding event^{1,17,18}, mutations that reduce the catalytic efficiency of the deaminase domain of a base editor might prevent it from catalyzing successive rounds of deamination before dissociating or being displaced from DNA. Once any C:G to T:A editing event has taken place, the sgRNA no longer perfectly matches the target DNA sequence and re-binding of the base editor to the target locus should be less favorable. Alternatively, mutations to the deaminase domain might alter substrate binding, the conformation of bound DNA, or substrate accessibility to the active site in ways that reduce tolerance for non-optimal presentation of a C to the deaminase active site. We sought to test both possibilities to discover new base editors that distinguish among multiple cytidines within the original editing window.

Given the absence of an available APOBEC1 structure, we identified mutations previously reported to modulate the catalytic activity of APOBEC3G, a cytidine deaminase that shares 42% sequence similarity of its active site-containing domain to that of APOBEC1 (ref. 19). We incorporated the corresponding APOBEC1 mutations into BE3 and evaluated their effect on base-editing efficiency and editing window width in HEK293T cells at two C-rich genomic sites containing Cs at positions 3, 4, 5, 6, 8, 9, 10, 12, 13, and 14 (site A); or containing Cs at positions 5, 6, 7, 8, 9, 10, 11, and 13 (site B). For analysis purposes, we define the “editing window width” as the number of nucleotide positions at a given site for which editing efficiency exceeds the half-maximal value for that target site. The editing window width of BE3 for the two C-rich genomic sites tested was 4 (site A) and 6 (site B) nucleotides (**Fig. 2a**).

Mutating residues essential for deaminase activity, such as R118A, led to dramatic losses of base-editing efficiency (**Supplementary Fig. 6**). Other mutations, however, narrowed the editing window while maintaining substantial editing efficiency (**Fig. 2a** and **Supplementary Fig. 6**). W90 in APOBEC1, corresponding to W285 in APOBEC3G, is predicted to be crucial for the formation of a hydrophobic active site in APOBEC1 (refs. 17,18). APOBEC3G mutant W285A has undetectable deaminase activity^{17,18}, and BE3 W90A similarly shows almost no base-editing activity (**Supplementary Fig. 6**). We hypothesized that W90Y or W90F might decrease the hydrophobicity of the active site while maintaining catalytic activity. Indeed, W90Y and W90F only modestly decreased base-editing activity while narrowing the editing window width at sites A and B to 3 or 2 nucleotides, respectively (**Fig. 2a** and **Supplementary Fig. 6**). These results demonstrate that mutations of the cytidine deaminase domain can narrow the width of the base-editing window.

Next we tested whether mutations to residues involved in APOBEC1 substrate binding could also alter the editing window width. R126 in APOBEC1 is predicted to interact with the phosphate backbone of single-stranded DNA¹⁸ and mutation of the corresponding resi-

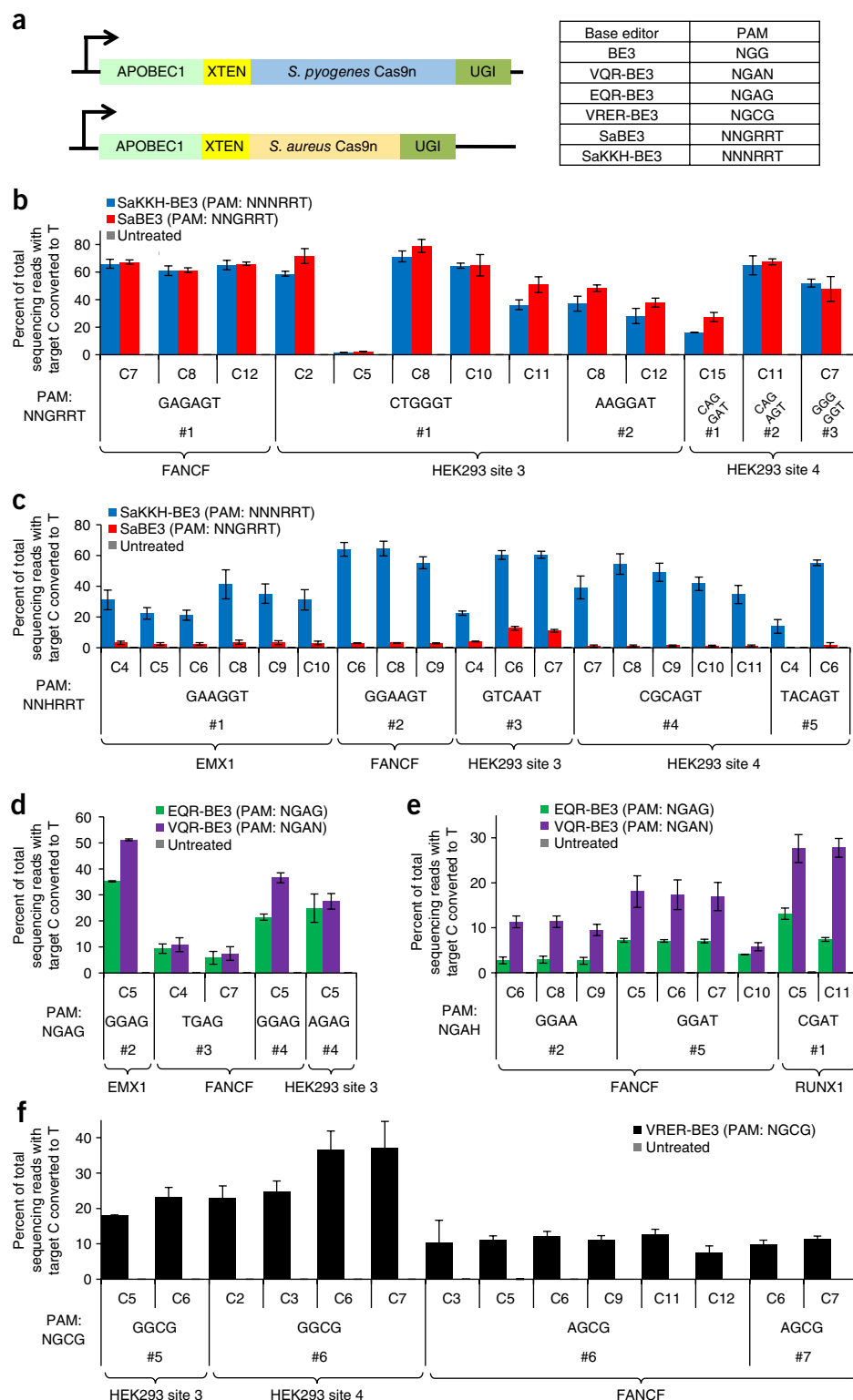


Figure 1 SaBE3, SaKKH-BE3, VQR-BE3, EQR-BE3, and VRER-BE3 mediate efficient base editing at target sites containing non-NGG PAMs in human cells. (a) Base editor architectures using *S. pyogenes* and *S. aureus* Cas9, and recently characterized Cas9 variants with alternate or relaxed PAM requirements^{15,16}. (b–f) HEK293T cells were treated with the base editor variants shown. The percentage of total DNA sequencing reads (with no enrichment for transfected cells) with C converted to T at the target positions indicated are shown. The PAM sequence of each target tested is shown below the x axis. See **Supplementary Sequences 1** for a list of full target sequences. The charts show the results for SaBE3 and SaKKH-BE3 at genomic loci with NNGRRT PAMs (b), SaBE3 and SaKKH-BE3 at genomic loci with NNHRRT PAMs (c), VQR-BE3 and EQR-BE3 at genomic loci with NGAG PAMs (d) and with NGAH PAMs (e), and VRER-BE3 at genomic loci with NGCG PAMs (f). Values and error bars reflect the mean and s.d. of two to four biological replicates performed on separate days.

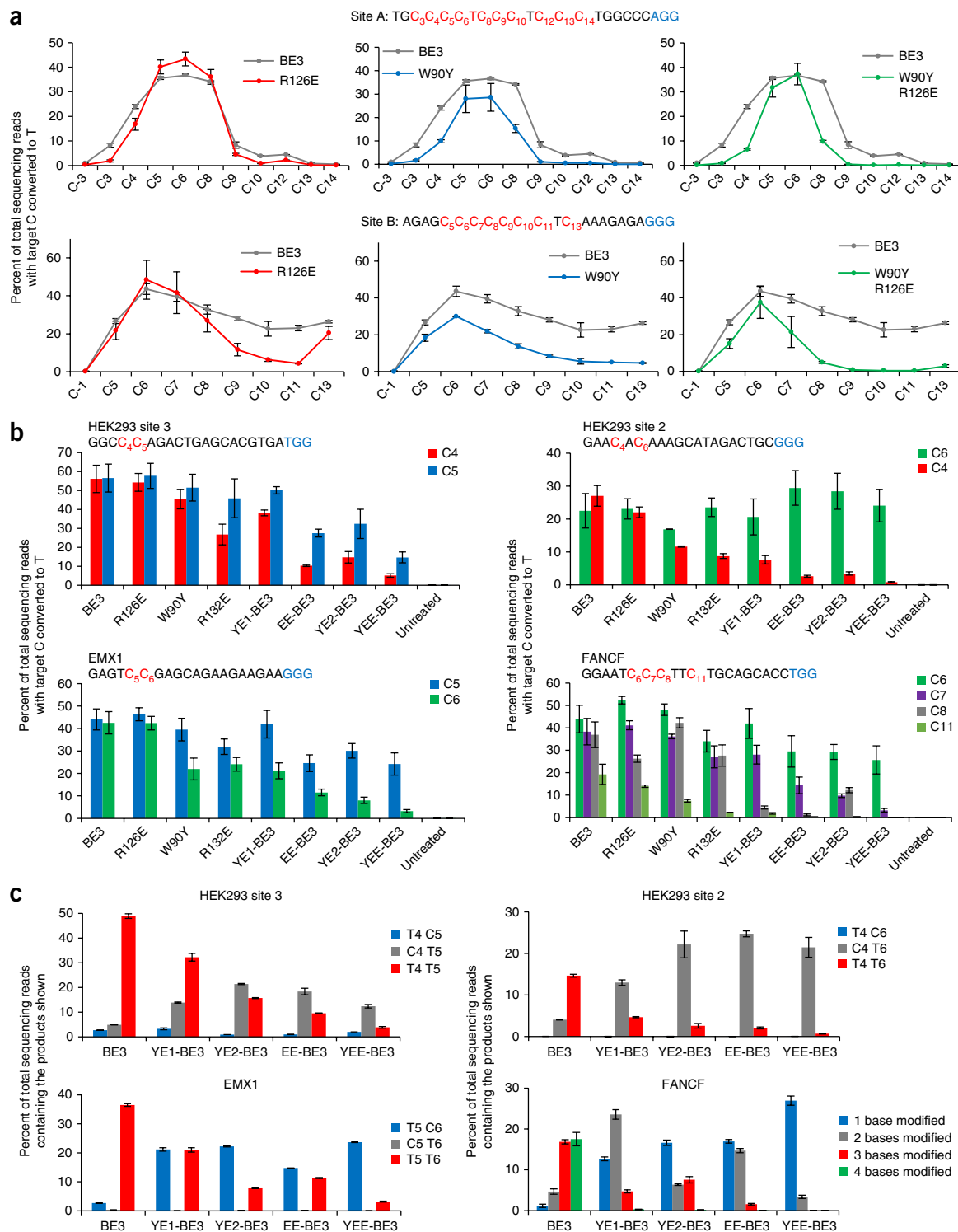


Figure 2 Base editors with mutant cytidine deaminase domains exhibit narrowed editing windows. **(a)** Cytidine deaminase mutations narrow the width of the editing window. **(b)** Effect of cytidine deaminase mutations that narrow editing window width on genomic loci containing multiple Cs within the canonical BE3 editing window (positions 4–8). Combining mutations has an additive effect on narrowing the editing window. YE1 = W90Y + R126E; EE = R126E + R132E; YE2 = W90Y + R132E; YEE = W90Y + R126E + R132E. **(c)** YE1-BE3, YE2-BE3, EE-BE3, and YEE-BE3 alter the product distribution of base editing, producing predominantly singly modified products, in contrast with BE3, even when multiple Cs are present in positions 4–8. Values and error bars reflect the mean and s.d. of two to four biological replicates performed on separate days.

due to Ala in APOBEC3G (R320A) decreases apparent activity by at least fivefold¹⁸. Notably, when introduced into APOBEC1 in BE3, R126A, and R126E maintained activity comparable to that of BE3 at the most strongly edited central positions (C5 and C6 for site A, C6 and C7 for site B), while decreasing editing activity at other positions (Fig. 2a and Supplementary Fig. 6). Both of these two mutations narrowed the width of the editing window at site A and site B to 3 nucleotides (Fig. 2a and Supplementary Fig. 6). R132 is near R126 and could also influence single-stranded DNA accessibility. R132E led to a decrease in editing efficiency but also narrowed the width of the editing window to 3 nucleotides for both sites A and B (Supplementary Fig. 6).

We combined W90Y/F, R126E, and R132E, the mutations that narrowed the editing window without drastically reducing base-editing activity, into doubly and triply mutated base editors. Combining the R126E and W90F mutations modestly narrowed the editing window compared to the W90F mutant, while combining W90F with R132E resulted in a dramatic loss of activity (Supplementary Fig. 6). The double mutant W90Y+R126E, however, resulted in a base editor (YE1-BE3) with BE3-like maximal editing efficiencies, but substantially narrowed editing window width of ~2 nucleotides for both site A and site B (Fig. 2a). The W90Y+R132E base editor (YE2-BE3) exhibited modestly lower editing efficiencies, but also narrowed the editing window width to ~2 nucleotides for both site A and site B (Supplementary Fig. 6). The R126E+R132E double mutant (EE-BE3) showed similar maximal editing efficiencies and editing window width as YE2-BE3 (Supplementary Fig. 6). The triple mutant W90Y+R126E+R132E (YEE-BE3) exhibited 2.9-fold lower average maximal editing yields but very little editing beyond the C6 position and an editing window width of ~2 nucleotides and 1 nucleotide for site A and site B, respectively (Supplementary Fig. 6). These data taken together indicate that mutations in the cytidine deaminase domain can strongly affect editing window widths, in some cases with minimal or only modest effects on editing efficiency.

Next we compared the base-editing outcomes of BE3, YE1-BE3, YE2-BE3, EE-BE3, and YEE-BE3 in HEK293T cells targeting four well-studied human genomic sites that contain multiple Cs within the BE3 activity window¹. These target loci contained target Cs at positions 4 and 5 (HEK293 site 3), positions 4 and 6 (HEK293 site 2), positions 5 and 6 (EMX1), or positions 6, 7, 8, and 11 (FANCF). BE3 exhibited little preference for editing any of the Cs within the activity window of positions 4–8 (<1.2-fold average base preference for HEK293 sites 2, 3, and EMX1). In contrast, the narrow-window base editors exhibited increasing preference for editing one cytidine over others. For example, YEE-BE3 showed 31-fold preference for editing C6 over C4 at HEK293 site 2 (Fig. 2b). We also assessed editing of YE1-BE3, EE-BE3, YE2-BE3, and YEE-BE3 in U2OS cells, and observed similar alteration of base-editing window widths (Supplementary Fig. 2b) as we observed in HEK293T cells. These findings establish that mutant base editors with narrowed editing windows can discriminate between adjacent Cs, even when both nucleotides are within the BE3 editing window.

We also analyzed the product distributions generated by each of the above base editors to assess their apparent processivity. BE3 generated predominantly T4-T5 (HEK site 3), T4-T6 (HEK site 2), and T5-T6 (EMX1) products in treated HEK293T cells, resulting in, on average, 7.4-fold more products containing two Ts, than products containing a single T (Fig. 2c). In contrast, YE1-BE3, YE2-BE3, EE-BE3, and YEE-BE3 showed substantially higher preferences for singly edited C4-T5, C4-T6, and T5-C6 products at these sites (Fig. 2c). The YEE-BE3 triple mutant strongly favored single-T products by an average of 14-fold across the three genomic loci (Fig. 2c). Interestingly, for the target site in which only one C is within the target window (HEK293

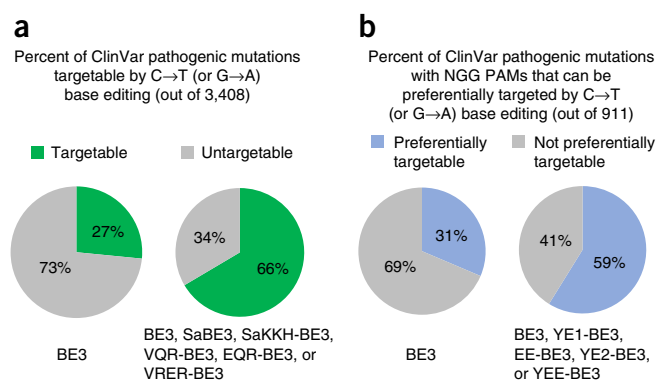


Figure 3 Genetic variants in ClinVar that can be corrected in principle by the base editors developed in this work. The NCBI ClinVar database of human genetic variations and their corresponding phenotypes⁵ was searched for genetic diseases that in theory can be corrected by base editing. **(a)** The base editors with altered PAM specificities developed in this study substantially increases targetable loci among all pathogenic T→C or A→G mutations in the ClinVar database. **(b)** Improvement in base-editing targeting scope among all pathogenic T→C or A→G mutations in the ClinVar database through the use of base editors with narrowed activity windows. BE3 is assumed to edit Cs in positions 4–8 with comparable efficiency, as shown in Figure 2. YE1-BE3, EE-BE3, YE2-BE3, and YEE-BE3 are assumed to edit with a preference of C5 > C6 > C7 ≈ C4. The blue fractions denote the proportion of pathogenic T→C or A→G mutations that can be edited by BE3 without comparable editing of other Cs (left), or that can be edited by BE3 or by one of BE3 mutants without comparable editing of other Cs (right).

site 4, at position C5), all four mutants exhibited editing efficiencies comparable to those of BE3 (Supplementary Fig. 7). These findings together suggest that the decreased apparent processivity of these narrow-window base editors favors conversion of only a single C at target sites containing multiple Cs within the BE3 editing window. These data also suggest a positional preference of C5 > C6 > C7 ≈ C4 for these mutant base editors, although we note that this preference could differ depending on the target sequence.

To test if a narrow-window base editor maintains its editing window profile over time following a single treatment, we treated HEK293T cells with YE1-BE3, EE-BE3, YE2-BE3, or YEE-BE3 and analyzed by HTS the distribution of base editing within the protospacer sequence 3 d, 6 d, or 9 d after treatment. We observed no substantial change in the positional distribution of base editing over 9 d at any of the four genomic loci tested (Supplementary Fig. 8).

Next, we examined the off-target activity of the base editors with narrowed activity windows. We assayed by HTS 21 known off-target loci corresponding to three on-target loci. We observed on average 3.6-fold less efficient off-target base editing per on-target editing event of the BE3 variants with narrowed activity windows when compared to BE3 across the 21 off-target sites (Supplementary Fig. 9). This improvement in DNA specificity may arise from a reduction in the residence time of base editors at off-target loci that further impedes deamination with catalytically impaired APOBEC1 domains.

Finally, we combined the window-modulating mutations in APOBEC1 with the Cas9 variants with different PAM specificities. For example, VQR-YE1-BE3 allowed editing with a narrowed activity window and greater positional selectivity of target sites containing an NGA PAM (Supplementary Fig. 10a). Not all window-narrowing mutations could be productively combined with altered PAM base editors, however. When window-narrowing mutations were installed in SaKKH-BE3, for example, we observed a decrease in base-editing

efficiency without any obvious change in the width of the activity window, perhaps arising from differences in the substrate accessibility of this base editor compared with that of BE3 and its variants (Supplementary Fig. 10b).

The five base editors with altered PAM specificities described in this study together increased by 2.5-fold the number of disease-associated mutations in the ClinVar database that can, in principle, be corrected by base editing (Fig. 3a). Similarly, the development of base editors with narrowed editing windows approximately doubled the fraction of ClinVar entries with a properly positioned NGG PAM that can be corrected by base editing without comparable modification of a non-target C (Fig. 3b).

These developments substantially expand the targeting scope of base editing by developing base editors that use Cas9 variants with different PAM specificities, and by developing a collection of deaminase mutants with varying editing window widths. The modularity of base editing established in this study suggests that the advances described here should also be applicable to other fusions of Cas9 variants linked to nucleotide conversion enzymes.

METHODS

Methods, including statements of data availability and any associated accession codes and references, are available in the [online version of the paper](#).

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

Y.B.K., A.C.K., J.M.L., and K.T.Z. conducted the experiments. M.S.P. performed computational analyses. D.R.L. supervised the research. All authors contributed to writing the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare competing financial interests: details are available in the [online version of the paper](#).

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ONLINE METHODS

Cloning. PCR was performed using Q5 Hot Start High-Fidelity DNA Polymerase (New England BioLabs). Plasmids for BE and sgRNA were constructed using USER cloning (New England BioLabs) from previously reported plasmids¹. See **Supplementary Sequences 2** for a full list of primers used in this study. DNA vector amplification was carried out using NEB 10beta competent cells (New England BioLabs). Site-directed mutagenesis of APOBEC1 variants was done using blunt-end ligation. Briefly, a primer with an overhang containing the desired point mutation was used to amplify the appropriate vector plasmid by PCR. KLD enzyme mix (New England BioLabs) was used to phosphorylate and circularize the PCR product before transformation.

Cell culture. HEK293T (ATCC CRL-3216) and U2OS (ATCC HTB-96) cells were cultured in Dulbecco's Modified Eagle's Medium plus GlutaMax (ThermoFisher) supplemented with 10% (v/v) FBS (FBS), at 37 °C with 5% CO₂. Cell lines have been validated by vendors, and were tested using the MycoAlert PLUS Mycoplasma Detection Kit (Lonza).

Transfections. HEK293T cells seeded on 48-well collagen-coated BioCoat plates (Corning) were transfected at ~70% confluency. 750 ng of BE and 250 ng of sgRNA expression plasmids were transfected using 1.5 µl of Lipofectamine 2000 (ThermoFisher Scientific) per well according to the manufacturer's protocol. 500 ng of BE and 250 ng of sgRNA expression plasmids were transfected into U2OS cells using a Lonza 4D-Nucleofector with the DN-100 program according to the manufacturer's protocols.

High-throughput DNA sequencing of genomic DNA samples. Transfected cells were harvested after 3 d. The genomic DNA was isolated using the Agencourt DNAdvance Genomic DNA Isolation Kit (Beckman Coulter) according to the manufacturer's instructions. Genomic regions of interest were amplified by PCR with flanking HTS primer pairs listed in the **Supplementary Sequences 2**. PCR amplification was carried out with Phusion hot-start II DNA polymerase (ThermoFisher) according to the manufacturer's instructions.

PCR products were purified using RapidTips (Diffinity Genomics). Secondary PCR was performed to attach sequencing adaptors. The products were gel-purified and quantified using the KAPA Library Quantification Kit-Illumina (KAPA Biosystems). Samples were sequenced on an Illumina MiSeq as previously described¹.

Data analysis. Nucleotide frequencies were assessed using a previously described MATLAB script¹. Briefly, the reads were aligned to the reference sequence via the Smith-Waterman algorithm. Base calls with Q-scores below 30 were replaced with a placeholder nucleotide (N). This quality threshold results in nucleotide frequencies with an expected theoretical error rate of 1 in 1,000.

Analyses of base-editing processivity were performed using a custom python script (**Supplementary Note 1**). This program trims sequencing reads to the 20 nucleotide protospacer sequence as determined by a perfect match for the 7 nucleotide sequences that should flank the target site. These targets were then consolidated and sorted by abundance to assess the frequency of base-editing products.

Bioinformatic analysis of the ClinVar database of human disease-associated mutations was performed in a manner similar to that previously described but with small adjustments (**Supplementary Note 2**)¹. These adjustments enable the identification of targets with PAMs of customizable length and sequence. In addition, this improved script includes a priority ranking of target C positions (C5 > C6 > C7 > C8 ≈ C4), thus enabling the identification of target sites in which the on-target C is either the only cytosine within the window, or is placed at a position with higher predicted editing efficiency than any non-target C within the editing window.

Data availability. Plasmids encoding SaBE3, SaKKH-BE3, VQR-BE3, EQR-BE3, VRER-BE3, YE1-BE3, EE-BE3, YE2-BE3, and YEE-BE3 are available from Addgene (plasmids 85169, 85170, 85171, 85172, 85173, 85174, 85175, 85176, 85177). See **Supplementary Sequences 3** for full amino acid sequences. High-throughput sequencing data have been deposited in the NCBI Sequence Read Archive database under accession code [SRP095823](https://www.ncbi.nlm.nih.gov/sra/SRP095823).